

DOCRED-FE: A DOCUMENT-LEVEL FINE-GRAINED ENTITY AND RELATION EXTRACTION DATASET

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ABSTRACT

Joint entity and relation extraction (JERE) is one of the most important tasks in information extraction. However, most existing works focus on sentence-level coarse-grained JERE, which have limitations in real-world scenarios. In this paper, we construct a large-scale document-level fine-grained JERE dataset DocRED-FE, which improves DocRED with Fine-Grained Entity Type. Specifically, we redesign a hierarchical entity type schema including 11 coarse-grained types and 119 fine-grained types, and then re-annotate DocRED manually according to this schema. Through comprehensive experiments we find that: (1) DocRED-FE is challenging to existing JERE models; (2) Our fine-grained entity types promote relation classification. We make DocRED-FE with instruction and the code for our baselines publicly available at <https://github.com/PKU-TANGENT/DOCRED-FE>.

Index Terms— Joint Entity and Relation Extraction, Information Extraction, Fine-Grained Entity Types

1. INTRODUCTION

The goal of joint entity and relation extraction (JERE) is to identify named entities and their relations from unstructured text. It is an essential problem in information extraction (IE) since it is critical to constructing knowledge graph. Recently, various works on JERE, including pipelined and end-to-end approaches, have achieved remarkable success [1, 2, 3].

Despite these successful efforts, most existing JERE works are conducted on sentence-level coarse-grained datasets, e.g., ACE04/05 [4], CoNLL04 [5], SciERC [6], TACRED [7], FewRel [8]. However, in real-world application scenarios, the model usually needs to understand the semantics of document-level texts and extract more precise information.

The research on document-level fine-grained entity and relation extraction requires a large-scale annotated dataset for both training and evaluation. It is usually expensive and time consuming to construct such a large-scale dataset from scratch. Therefore, we choose to re-annotate a popular document-level

Document

*Us and Them*_[Musical_work] is the fifth studio album by *English*_[Country] band *Godflesh*_[Musical_group]. It was released through *Earache Records*_[Record_label] on *17 May 1999*_[Time] in *Europe*_[Continent] and on *8 June 1999*_[Time] in *North America*_[Continent]. As with Godflesh's *1997*_[Time] remix album *Love and Hate in Dub*_[Musical_work], *Us and Them* is influenced by breakbeats, drum and bass, oldschool jungle, *trip hop*_[Style] and *hip hop*_[Style]. Shortly after *Us and Them*'s release, Godflesh frontman *Justin Broadrick*_[Musician] admitted that he "hated" the album as it was the expression of an "Identity crisis".

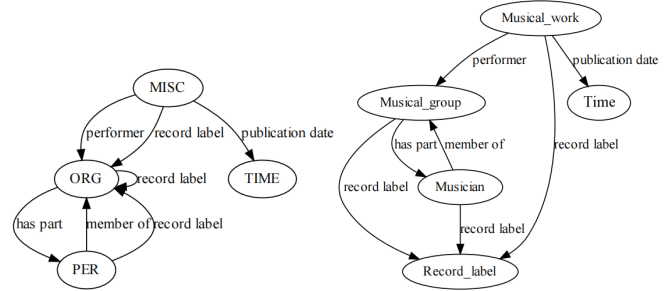


Fig. 1: A document case of DocRED-FE, with the corresponding origin (left) and new (right) schema. Only highlights the first mention of each entity.

relation extraction dataset, DocRED [9]. There are 96 fine-grained relation types in DocRED, while the entities are divided into only 6 types and are recognized by automatic tools.

In this paper, we propose DocRED-FE, a large-scale human-annotated fine-grained entity and relation extraction dataset based on DocRED. Specifically, we first re-design the entity type schema to a hierarchical structure including 11 coarse-grained types and 119 fine-grained types. Combined with the 96 fine-grained relation types in original DocRED, DocRED-FE contains richer contextual information with a finer granularity. We illustrate an example document with its corresponding schema in Fig. 1. Compared to the coarse schema in DocRED, our newly designed entity schema is more precise and expressive with most of the self-pointing relation edges dissolved, such as *ORG-record_label-ORG*.

To assess the JERE benchmark on DocRED-FE, we adopt

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several representative document-level JERE baselines, including a pipeline model based on BERT [10] and an end-to-end model based on BART [11]. Experimental results show that document-level fine-grained JERE poses a great challenge to current models and remains an open problem. Furthermore, we also confirm that fine-grained entity information can improve the performance on relation classification (RC). We believe our analysis can help to design more powerful models.

2. DATASET CONSTRUCTION

DocRED-FE is built based on DocRED, a large-scale crowd-sourced dataset from Wikipedia and Wikidata. DocRED is mainly built for relation extraction and has only 6 coarse entity types. Thus, we develop DocRED-FE via refining the entity type schema and re-annotating the entities in DocRED.

2.1. Schema Design

In this section, we introduce our *bottom-up* data-driven annotation approach. We aim to establish a schema which can reflect the characteristics of entities in texts properly, and make the frequency of each label in schema as balanced as possible. Therefore, controlling the granularity and dividing boundaries of the schema is particularly critical. Formally, we design the entity schema in the following three steps:

Stage 1: Entity Linking. For each entity in the dataset, we link it to Wikidata to get its types. For example, for the entity *China*, we get its candidate types as *Country*, *State*, *Socialist State*. Then we statistically compute the frequency of each type and filter about top-100 of them to constitute our initial fine-grained candidate set.

Stage 2: Combining with Existing Schema. As for FIGER [12] and Few-NERD [13], they are both general encyclopedia datasets. Considering their homology with DocRED, we combine their entity schema with our initial candidate set. In the process of combination, both coarse-grained and fine-grained types are adjusted. For example, we prefer to make the initial fine-grained type *GPE* in Few-NERD to be coarse-grained one, while for the types *Military_operation*, *War*, *Battle* that have similar meaning, we only keep the first one remained. After the prior two stages, a preliminary hierarchy schema is built.

Stage 3: Iterative Exploratory Annotation and Refinement. To validate whether this schema is suitable, we carry out a few rounds of exploratory annotation, and refine the schema according to the feedback by the annotators after each round. For example, we subdivide the type *GPE* into more subtypes not only *Country*, *City*, but also *Continent*, *State&Province* since entities with type *GPE* account for a large proportion. We also merge type *Actor* and *Director* into *Actor&Director* since they are always used to label the same person and hard to distinguish. Consequently, the finalized schema includes 11 coarse-grained types and 119 fine-grained types is established.

Table 1: Comparison to well-known RE/JERE datasets (Doc.: document, Tok.: token, Ent.: entity, ET.: entity type, Rel.: relation, RT.: relation type).

Dataset	# Doc.	# Tok.	# Ent.	# ET.	# Rel.	# RT.
ACE2005	-	259k	37,622	51	7,786	18
TACRED	-	3,866k	212,528	17	106,264	41
SciERC	-	65,334	1,015	6	2,687	7
FewRel	-	1,397k	112,000	-	56,000	100
DWIE	802	501k	23,130	311	21,749	65
DocRED	5,053	1,002k	98,610	6	56,354	96
DocRED-FE	2,596	516k	50,549	119	32,366	96

2.2. Human Annotation

In this stage, we perform human re-annotation on the entities according to our new schema. To ease the annotation burden, we ask annotators to assign one single type that best suits the entity according to its contexts. For example, sometimes *World War II* represents a book but not the war, so it should be tagged by *Written_work* rather than *Military_operation*. Sometimes a person is a *Soldier*, *Politician* and *Artist*, we ask the annotators to select the most prominent type depending on the contexts.

Annotators consist of about 20 experts that work on NLP research. They have linguistic knowledge and are instructed with detailed and formal annotation rules. To ease the annotation, the description of each entity on Wikipedia and its structured information on Wikidata are provided to annotators as references. We ensure that all the annotators are fairly compensated by market price according to their workload. The dataset is randomly divided and delivered to different annotators. To ensure the quality of our dataset, we conduct consistency checking via calculating the Cohen’s Kappa Score [14], and the result is 68.58%, which indicates a relatively high degree of consistency.

3. DATA ANALYSIS

In this section, we analyze our dataset from various aspects to have a deeper understanding.

3.1. Comparison with Related Datasets

Without the test part, DocRED-FE consists of 65% of the train part and whole dev part of the DocRED. Table 1 shows statistics of our dataset and some well-known RE/JERE datasets. Except DWIE [15], all of above are only fine-grained in either entity or relation. As for DWIE, we do not have enough statistics on entity information since they do not provide a complete entity schema. To the best of our knowledge, it has fewer documents and tokens but more complicated entity types, which make NER difficult due to insufficient training.

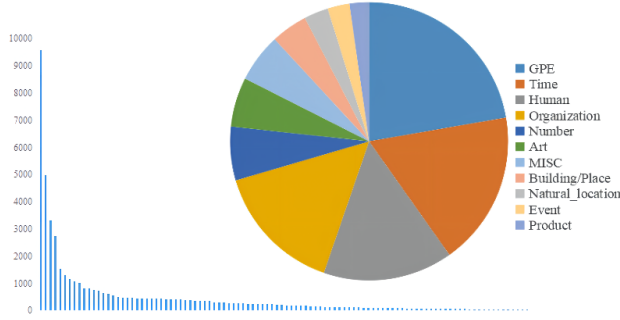


Fig. 2: Statistics on first-level (pie chart) and second-level (bar chart) entity type.

3.2. Entity Distribution

In Figure 2, the pie chart shows the entity distribution of first-level entity type while the bar chart shows that of the second-level. For the bar chart, the top-5 types are *Time*, *Country*, *Number*, *City* and *Musician*, the last-5 are *Chemical_and_biological*, *Natural_phenomenon*, *Dam*, *Disease* and *Sports_season*. Long-tail phenomenon still exists, but the downtrend is relatively flat after the top-5.

3.3. Schema for Each Document

After our transformation from the original coarse schema to the finer one, the triplets in our schema also differ from DocRED due to the variety of the head and tail entity type as shown in Figure 1. Compared with the latter, there are following optimizations: (1) Diversity of the entity type makes the relation distribution more even, amount of self-pointing edge descends, and overall density tends to be more reasonable. (2) Naming consistency of entity and relation makes the meaning clear and unambiguous, domain information like music becomes prominent. (3) The rich entity information can provide more significant hints for RC tasks. To measure the gain quantitatively, we compute the information gain ΔH and the information gain ratio ΔH_{ratio} comparing to the original DocRED. The information entropy means average weighted information entropy of the triplet set computed by each entity type pair [16], the ΔH means the value of decline of the information entropy between DocRED and DocRED-FE, the result is 1.09. The ΔH_{ratio} is 65.67%, which means the ΔH relative to the information entropy of DocRED. It theoretically indicating that fine-grained entity types can benefit RC.

4. EXPERIMENTS

We conduct two experiments using the proposed DocRED-FE dataset. Firstly, in Section 4.1, we conduct a JERE experiment on the DocRED and DocRED-FE to show that DocRED-FE provides a more difficult benchmark. Secondly, in Section 4.2, we show that the DocRED-FE can help the RC by providing double-level and fine-grained entity type information.

Table 2: Performance of different JERE models on DocRED and DocRED-FE (relations).

Model	DocRED		
	RE F1 (strict)	RE F1 (relaxed)	NER F1
JEREX	40.26	40.62	80.25
REBEL	45.29	46.45	-

Model	DocRED-FE		
	RE F1 (strict)	RE F1 (relaxed)	NER F1
JEREX	31.52	39.02	67.74
REBEL	37.51	45.94	-

For the JERE task, we adopt two JERE models as baselines of our dataset, including a pipelined model JEREX [17] and an end-to-end generation model REBEL [18]. JEREX links entity recognition and relation classification tasks by feeding the results of entity recognition to the subsequent relation classification. REBEL employs BART-large as the base model and translates a raw input dataset into a set of triplets that can be uniquely decoded into entity-relation pairs.

For the RC task, we adopt several representative RC models, including CNN, LSTM, BiLSTM proposed from DocRED, GAIN [19], JEREX and SSAN [20].

For both experiments, we train the model using the same part of DocRED and DocRED-FE. As mentioned above, we use 21k entity-relation pair instances as the train part and 10k instances as the validation part. We use the default settings mentioned in the references and fine-tune the model until the loss function converges and report the results on the validation part.

4.1. Joint Entity and Relation Extraction

In this experiment, we follow previous work [17, 21] and use two modes of F1 score. The strict mode counts a relation as correct if and only if its mention span, coreference resolution, entity type and relation type are correct. The relaxed mode focuses on the relation accuracy and ignores the correctness of the entity type. When acquiring the golden entity type during pipelined process, 6 entity types in DocRED are replaced by 119 entity types when testing DocRED-FE.

As shown in Table 2, both the strict and relaxed F1 of a model trained on DocRED-FE are lower than those trained on DocRED. Strict F1 declines as expected due to the more complex entity types. The reason why relax F1 become slightly lower is that although fine-grained entities provide more information, incorrect entity predictions may have a negative impact on relation classification and degrade the whole JERE performance. The result suggests that DocRED-FE presents great challenges to current models.

4.2. Relation Classification

While JERE task on DocRED-FE poses a challenge for existing models, we want to find how our double-level entity types help the RC part of JERE task and provide some reasonable methods. Therefore, we utilize different levels of entity information to build the following three methods:

1. Only use the fine-grained entity type id.
2. Use the fine-grained entity type id and its semantic representation.
3. Use the fine-grained entity type id and corresponding coarse-grained entity type id.

Only Fine-Grained Types Similar to Section 4.1, for the entity type embedding part in every models, we replace the original coarse-grained types id embedding by our fine-grained types id embedding. As shown in Table 3, fine-grained entities add more information to relation classifier and constrain the appropriate relation types. This helps the model to classify relations better.

Fine-Grained Types with Its Semantic Representation

While the entity type id provides necessary information for relation classification, the entity type semantic information could also be utilized. From this perspective, we use BERT to encode each entity type such as *Musician*, *Country*, *Military_operation* to get its 768-dim semantic embedding e_s . In the way of aggregating, we use concatenation or directly adding depending on different models.

For JEREX, we use MLP to map the e_s into target embedding shape, and concatenate the semantic embedding e_s , fine-grained type id embedding e_f with the entity-pair embedding e_{en} before feeding to the final relation classifier RC . This process can be expressed as the following formula.

$$Score = \mathbb{RC}[concat(e_{en}, e_f, e_s)] \quad (1)$$

For BiLSTM, GAIN, SSAN, we use MLP to map the e_s into the same shape with word embedding e_w , so that we can add these embeddings directly in the embedding block which is in the front of the model. By the way, we add "Padding" type for non-entity token intentionally. The token embedding E_t of SSAN compute as

$$E_t = e_w + e_f + e_s * \alpha \quad (2)$$

As shown in Table 4, model with added semantic information performs better than that without it. F1 of BiLSTM using semantic information improves from 49.28 to 50.26, indicating that adding semantic information in the DocRED-FE dataset is helpful to relation classification.

Both Fine-Grained and Coarse-Grained Types Considering the information in the hierarchical structure of entity schema of DocRED-FE, we further integrate new coarse-grained type based on first method. For simplicity, we train a coarse-grained type id embedding layer, project the new coarse-grained type id to type embedding e_c , and make fusion like the second method.

Table 3: Infusing different granularity of entity information into RC models. "Coarse-grained Type" represent the entity type in DocRED. We report the F1 score on dev set.

Model	Coarse-grained Type	Fine-grained Type
CNN	43.90	44.59
LSTM	47.72	49.34
BiLSTM	48.64	49.28
GAIN	58.09	58.35
JEREX	57.99	58.19
SSAN	58.10	58.24

Table 4: F1 represents only using fine-grained type id. F1 with SE represents additionally aggregating semantic representations of fine-grained types. F1 with DLE represents aggregating double-level entity information.

Model	F1	F1 with SE	F1 with DLE
BiLSTM	49.28	50.26	49.81
GAIN	58.35	58.69	58.61
JEREX	58.19	58.26	58.34
SSAN	58.24	58.75	58.30

For BiLSTM, GAIN and JEREX, concatenation operation as

$$Score = \mathbb{RC}[concat(e_{en}, e_f, e_c)] \quad (3)$$

For SSAN, the token embedding E_t compute as

$$E_t = e_w + (e_f + e_c) * \gamma \quad (4)$$

The second and fourth columns of Table 4 show that by adding double-level entity type information, RC models could utilize coarse-grained and fine-grained entity type jointly, and perform better than only using fine-grained entity type. That means, the hierarchical entity types could be leveraged in designing RC models.

5. CONCLUSION

In this paper, we introduce DocRED-FE, a new dataset consisting of double-level entity types, including 11 coarse-grained types and 119 fine-grained types. Compared with existing JERE dataset, DocRED, our proposed dataset provides a more difficult benchmark for current JERE models and richer entity-level information. To explore the advantages of our new entity schema, we propose and experiment some simple, general but effective ways. We hope DocRED-FE can contribute to future research on building accurate and robust JERE models.

6. ACKNOWLEDGEMENT

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